9	(a)(i)	$m = 235 \times 1.66 \times 10^{-27} = 3.90 \times 10^{-25}$ kg	A1
	(a)(ii)	Volume $v = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times \left(8.9 \times 10^{-15}\right)^3 = 2.95 \times 10^{-42}$ m^3 Density = $\frac{m}{v} = \frac{3.90 \times 10^{-25}}{2.95 \times 10^{-42}} = 1.32 \times 10^{17}$	B1 A1
	(a)(iii)	Since the density of the lump of uranium is a very small fraction of the density of the nucleus, (need to show comparison) this means that there are lots of empty space in the atom. / volume of <u>atom</u> is very large compared to volume of <u>nucleus</u> / mass is concentrated in a very small region (nucleus).	B1 B1
	(b)(i)	A nuclear reaction in which a <u>heavy nucleus splits</u> into two or more lighter nuclei, with release of energy.	B1 B1
	(b)(ii)	**1.** Binding energy of product particles $= 144 \times 1.3341 \times 10^{-12} + 90 \times 1.3864 \times 10^{-12}$ $= 3.169 \times 10^{-10}$ J Binding energy of reactant particles $= 235 \times 1.2191 \times 10^{-12}$ $= 2.865 \times 10^{-10}$ J Energy released = $3.169 \times 10^{-10} - 2.865 \times 10^{-10} = 3.04 \times 10^{-11}$ J	B1 B1 A1
	(b)(ii)	**2.** mass equivalent = $\frac{\text{energy released}}{c^2} = \frac{3.04 \times 10^{-11}}{\left(3.00 \times 10^8\right)^2}$ $= 3.38 \times 10^{-28}$ kg	B1 A1
	(b)(iii)	(The neutrons are constituent particles/free unbounded/single particles,) hence <u>there is no binding energy for the neutrons.</u>	B1
	(c)(i)	Decay constant is <u>probability per unit time of decay</u> of a radioactive <u>nucleus</u>.	**A1**
	(c)(ii)1.	$\lambda = \frac{\ln 2}{28.0 \times 365 \times 24 \times 60 \times 60} = 7.85 \times 10^{-10}$ s^{-1}	**A1**

	(c)(ii)2.	$A = \lambda N \Rightarrow N = \frac{6.4\times10^{9}}{7.8\times10^{-10}} = 8.21\times10^{18}$ mass of sample = $8.21\times10^{18}\times90\times1.66\times10^{-27}$ $= 1.23\times10^{-6}$ kg $= 1.2\times10^{-3}$ g	**C1** **C1** **A1**
	(c)(ii)3.	volume = $\frac{1.23\times10^{-3}}{2.54} = 4.8\times10^{-4}$ cm^3	**A1**
	(c)(iii)	The volume calculated in **(c)(ii)3.** is a very small volume of Strontium-90. It shows that this small volume of Strontium-90 already has high activity (10^9 Bq) compared to background count (10^{-1} Bq). Inhaling in dust particles exposes one to high radiation.	**B1**